UTM UNIVERSITI TEKNOLOGI MALAYSIA | Faculty of Computing

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Lecturer	:	DR. SHAMINI
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Prepared by :

Matric No.	Student Name
B23CS0031	AZHEEM SYAHMEE BIN ZULKIFLI

Assessment Mode : Individual / Group

Feedback from Lecturer :

Student Declaration

I hereby declare that this assessment is my own original work. I have not plagiarised any sources and have not used generative AI tools unless explicitly permitted by the lecturer. All references used have been properly acknowledged.

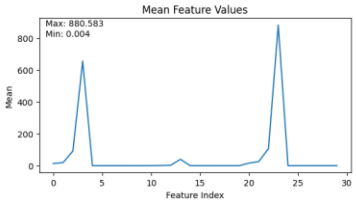
Signatures:
AZHEEM SYAHMEE

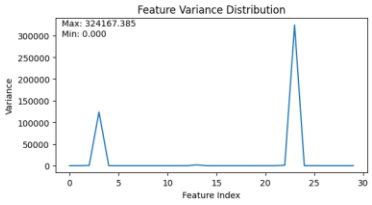
Data Preparation and Exploration

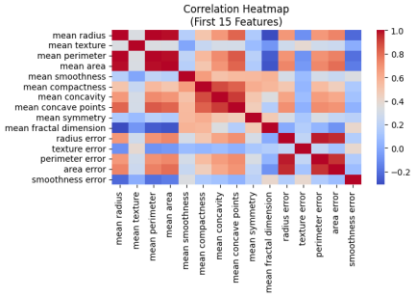
Breast cancer dataset was used as the source of biomedical data. It contains 569 samples and 30 features. The dataset loaded by using sklearn.datasets.load_breast_cancer().

ReactomePathways_2023-03-06.gmt dataset is used to simulate the pathway-level activity. There are total of Reactome pathways: 2593. 50 pathways is selected to use for generating synthetic pathway activity profiles.

After the dataset is loaded, we standardized it using StandardScaler to produce zero-mean which is used to improve training stability.







GAN Model Development

GAN is used to generate synthetic pathway activity profiles. The correct implementation of GAN is consisting of Generator which takes random noise (latent vector) as input and generates fake biomedical samples and Discriminator which is used to classified whether its fake or real.

GAN Architecture and Training Settings

Component	Configuration Used
Input Features (input_dim)	50 pathway features
Latent Dimension (latent_dim)	11
Hidden Dimension (hidden_dim)	128
Generator, Discriminator Activation	ReLU
Optimizer	Adam
Learning Rate (lr)	0.0001
Loss Function	BCELoss
Batch Size	569 samples
Epochs During Tuning	50
Final Training Epochs	100
Optuna Trials	10

Optuna and Grid Hyperparameter Tuning

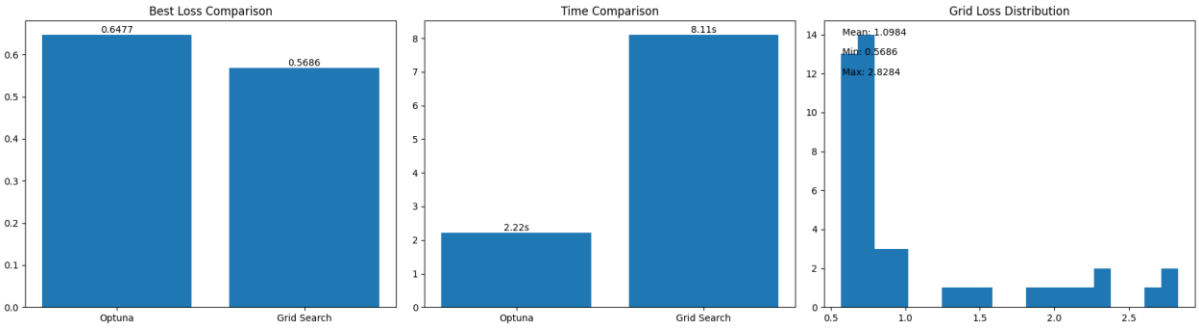
Hyperparameter tuning is used to optimize the GAN training parameter to improve its performance and accuracy. Its objective is used to minimize the loss so it able to learn more effectively and able to create a more quality fake data.

Hyperparameters Used

Hyperparameter	Search Space	Purpose
latent_dim	10 – 50	Size of random noise vector
hidden_dim	32, 64, 128	Number of hidden neurons
learning_rate (lr)	0.0001 – 0.01	Controls GAN learning speed

Training Configuration

Setting	Value
Number of Trials	10
Epochs per Trial	50
Optimizer	Adam
Loss Function	BCELoss



Best Hyperparameter Results Comparison

Parameter	Optuna Result	Grid Search Result
latent_dim	11	20
hidden_dim	128	64
learning_rate	0.0001	0.01
Best Generator Loss	0.6477	0.5686
Discriminator Loss	-	1.2865
Optimization Time	2.216 seconds	8.113 seconds

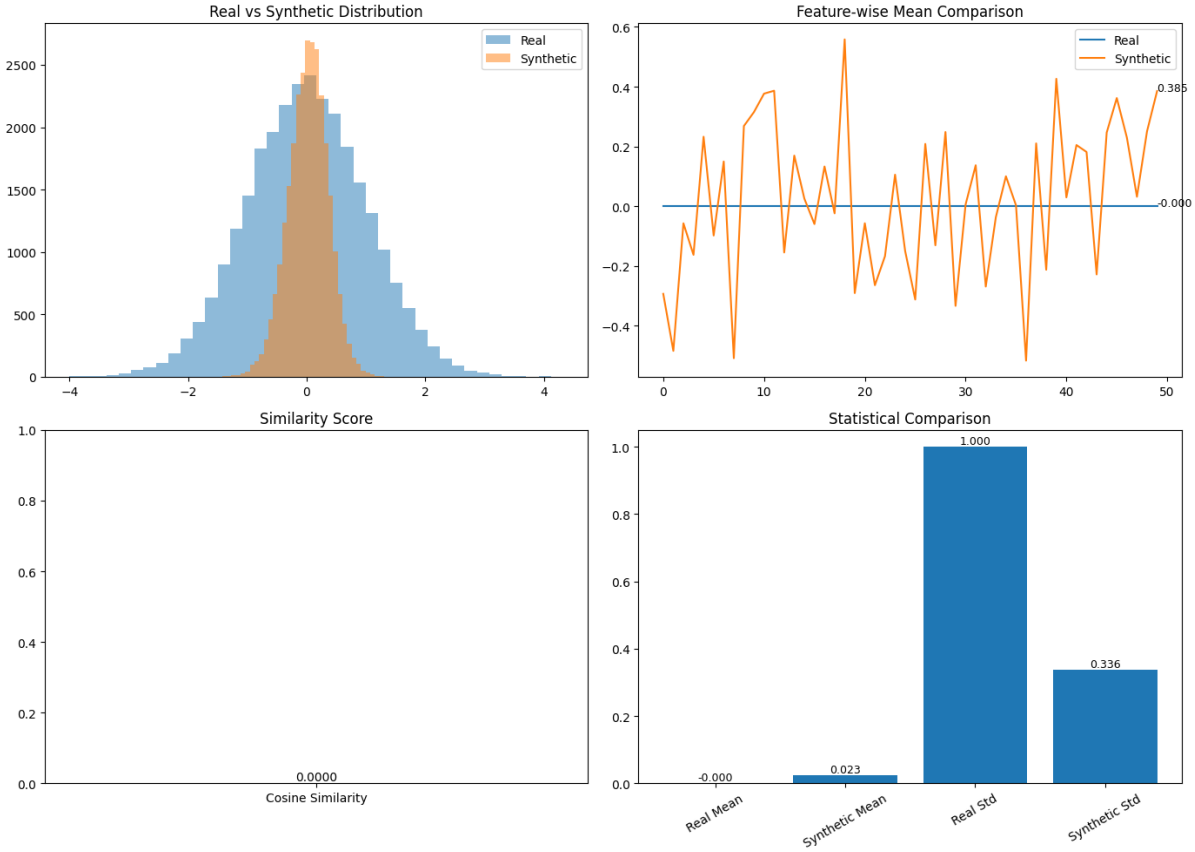
Optuna Result Summary

Based on the result we able to see that the lower learning_rate produced a more stable training. The hidden layer of 128 in this case is the best which able to improved Generator leaning capability and a small latent dimension reduced the loss occur in this tuning.

Grid Search Result Summary

This combination of hyperparameter provide the lowest generator loss probably because it able to achieve a great balance between their representation capacity and training stability. Most of the parameter is on middle ground not too smallest not too big which is probably just nice in this case.

Evaluation of Synthetic Data & critical analysis



Statistical Summary: Real vs Synthetic Data

Metric	Value
Real Mean	-5.244779e-18
Synthetic Mean	2.340562e-02
Real Std	1.000000
Synthetic Std	3.361617e-01
Cosine Similarity	1.631655e-17

Based on the distribution chart, it shows that some of the value overlap between real and synthetic which mean there exist value that are similar to real value.

For feature wise comparison chart, show the overall shape where the synthetic data try to match the real value but fail to be able to precisely match individual feature.

The cosine value of this experimentation is extremely close to zero which means that overall feature relationship between real and synthetic data are weak. This mean that GAN only able to catch certain learning pattern rather than full.

Overall, GAN method able to create synthetic data that almost similar to real dataset but fail to fully replicate it which shows there is improvement can be make. Some of improvement that can be make is increasing training epoch, using deeper GAN architecture and many more.

Best Overall Method: Optuna

Even though Grid Search method able to get a lower Generator loss, Optuna able to get similar Generator loss with less time compared to Grid search with a better overall tuning method for this experiment. Lower time taken is also equal to lower computational cost which what it means by Optuna able to get better overall tuning as it has a better balance between optimization, computational cost and performance.

Marking Rubric

Component	Excellent (5)	Good (4)	Satisfactory (2–3)	Poor (0–1)	Marks
Part 1: Data Preparation and Exploration	Complete preprocessing, normalization, visualization, and clear explanation of dataset and pathway features	Most preprocessing and visualization completed with minor issues	Basic preprocessing and limited explanation provided	Incomplete preprocessing or incorrect dataset handling	/5
Part 2: GAN Model Development	Correct GAN implementation with clear Generator and Discriminator explanation, proper training, and meaningful synthetic outputs	Functional GAN with minor implementation or explanation issues	Basic GAN implementation with limited understanding	Incorrect or incomplete GAN implementation	/5
Part 3: Optuna Hyperparameter Tuning	Correctly applies Optuna tuning with clear optimization process, parameter selection, and best parameter analysis	Good Optuna implementation with minor missing explanations	Basic Optuna tuning with limited understanding	No meaningful Optuna tuning performed	/5
Part 4: Grid Search Hyperparameter Tuning Comparison	Clearly compares Optuna and Grid Search with performance analysis, parameter comparison, and proper discussion	Comparison mostly correct but lacks detailed analysis	Basic comparison with limited evaluation	No proper comparison between tuning methods	/5
Component	**Excellent (10)**	**Good (6-8)**	**Satisfactory (5–4)**	**Poor (0–3)**	**Marks**
Part 5: Evaluation of Synthetic Data & critical analysis	Strong evaluation using suitable visualizations/statistical analysis with insightful interpretation of similarity between real and synthetic data	Good evaluation with minor limitations in analysis	Basic evaluation with limited interpretation	No proper evaluation or incorrect analysis	/10